

Date: _____

Day: _____

Normalization:-

its also performance improving of a model.

→ Let's take classification problem, in which age, salary, & output col.

→ it shows irregularity or inconsistency.

→ In unnormalized form loss function will be not symmetrical.



unnormalize



normalize

→ Simply we have to do feature scaling.

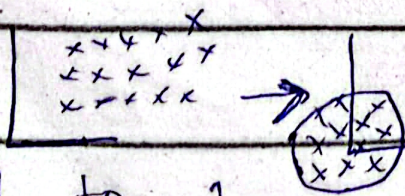
Standardize

Normalize.

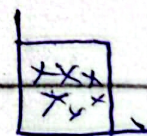
$$\text{value} \leftarrow \frac{x_i - \mu}{\sigma}$$

St. deviation $\leftarrow \sigma$

$$\frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$$



-1 to 1



0 to 1.

Date: _____

Day: _____

→ If u already know ^{value} that your column
max value or min value how much $\rightarrow$ normalize.
else standardize.

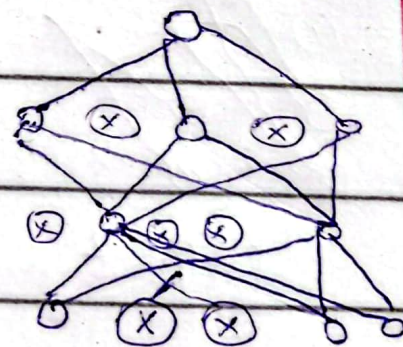
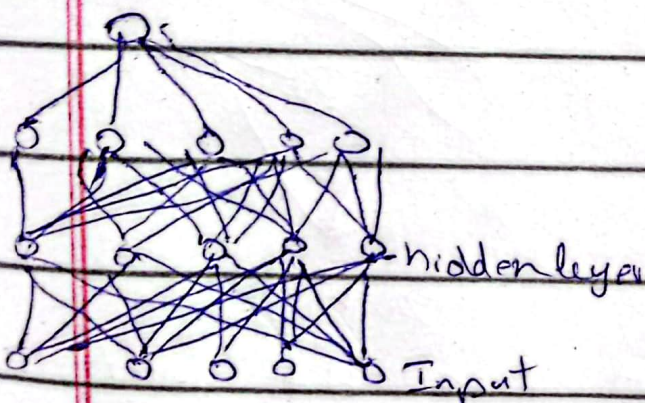
Dropout Layers:-

Problem of overfitting.

Possible sol:-

- 1) Add more data
- 2) Reduce complexity
- 3) Early stopping.
- 4) Regularization.
- 5) Dropout.

Dropout:-



a) Stand. Neural .Net

→ Randomly input & hidden layers nodes drop.

$p = 0.5$
↓
Dropout ratio.

Keep in 0.25 to 0.5
not to underfit also.

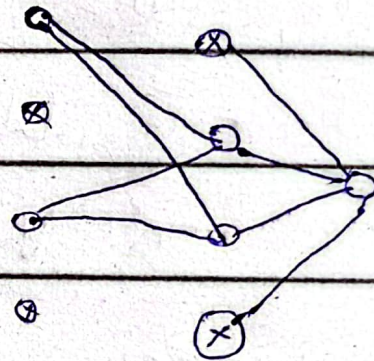
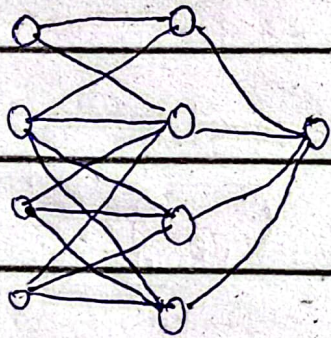
Date: _____

Day: _____

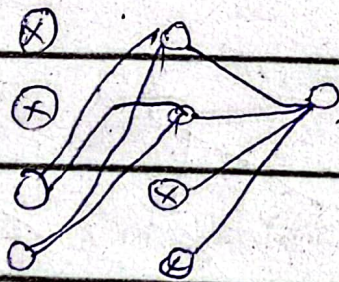
Let's we run 10 epochs our model.

So for every epoch switch randomly some nodes in input & hidden layer.

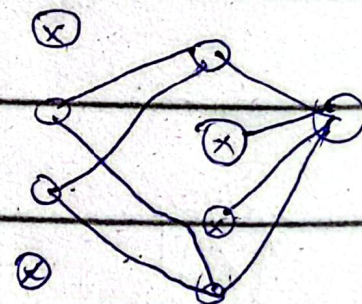
→ Actually u do training on 10 different N. Networks.



Epoch 1



Epoch 2



Epoch 3

When u take input from diff layers
so model don't try to focus just 1 point

Works like random forest algorithm.

Date: _____

Day: _____

R.F : 100 decision trees. $x_1, x_2, \dots, 1Y$

col sampling 60 % $\rightarrow 1$ 40 % 0.

$\rightarrow$ Same here u v making ensemble of nodes

$$\begin{aligned} w(1-p) \\ w \times 0.75 \end{aligned}$$

$\rightarrow$ Overfitting $P \uparrow$ underfitting $P \downarrow$

$\rightarrow$ Last layer $\rightarrow$ dropout

0	0	0
0	0	0
0	0	0

$\swarrow$

$\rightarrow$ CNN $\rightarrow$ 40-50 % $P \uparrow$

20-30 KNN, 10-50 ANN

Drawbacks: Convergence $\rightarrow$ delay.

Map:- Improving N.net model.

1) Overfitting:-

- Dropout layers
- Early stopping.
- Regularization.

2) Normalization:-

3) Van. gradient

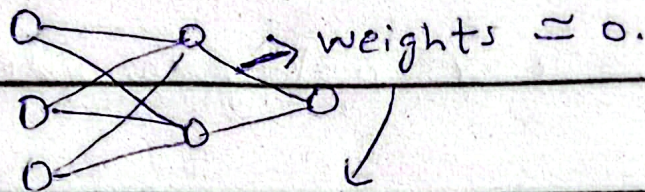
Don't add bias, added will always be weights

Date: _____

Day: _____

Regularization:-

Works to decrease overfitting.



it will be a kind of non existence.

complex model changes simple. $\left. \begin{matrix} \text{ } \\ \text{ } \end{matrix} \right\} \times$

Ways to solve overfitting:-

Adding more data

Reducing complexity model.

→ Add more rows.

→ Dropout.

→ Data argumentation.

→ early stopping

fake data adding

→ Regularization.

$\left[\begin{matrix} L_1 \\ L_2 \\ L_1 + L_2 \end{matrix} \right]$

Regularization:-

ANN find weight/biases

→ min Loss func

$L = \text{mse} \rightarrow \text{Reg.}$

$$C = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$$

→ binary cross entropy → cross

Have to minimize this.

$$L_1 = \frac{1}{2n} \sum \|w_i\|$$

$$L_2 = \frac{1}{2n} \sum \|w_i\|^2$$

Date: _____

99% we use L_2 .

Day: _____

In regularization we just add a
penalty term

$$C = L + \left[\frac{\lambda}{2n} \sum_{i=1}^K \|w_i\|^2 \right] \rightarrow \text{penalty}$$

$$w_1 \rightarrow w_{10}$$

$$\frac{1}{2n} [w_1^2 \dots w_{10}^2]$$

$\lambda \rightarrow$ hyperparameter.